

Raj Singh

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#) | [LeetCode](#)

SUMMARY

Final-year CS undergrad (CGPA: 7.2/10) seeking an AI/ML internship. Deployed 3 production AI systems with 100+ real users, including fine-tuned BERT models (91% accuracy) and LLM-powered applications. Skilled in end-to-end Machine Learning pipelines — model fine-tuning, RAG-based retrieval, Agentic AI workflows, and full-stack deployment using LangChain, HuggingFace, OpenAI, FastAPI, and GCP.

EXPERIENCE

My Trip Advisor | *Blithe Boom Startup* | [Live](#) Feb 2026 – May 2026
AI/Full-Stack Developer

- Built a GPT-4o powered travel planner in Next.js generating personalized itineraries in under 15 seconds, serving 100+ active users.
- Designed a multi-step LangChain prompt chaining pipeline for route optimization and budget planning, reducing manual effort by 70%.
- Integrated OpenAI Streaming API with JWT authentication and dynamic PDF export; handled 100+ itinerary generations with zero downtime.

PROJECTS

AI-Powered Cyber Threat Detection System Nov 2025 – Jan 2026

NLP | BERT Fine-Tuning | Flask | Flutter | Firebase | [GitHub](#)

- Fine-tuned BERT (HuggingFace Transformers) on 15,000+ labeled samples; achieved **91% classification accuracy** for phishing and harassment detection.
- Built a real-time inference pipeline using Flask; integrated with a Flutter mobile app achieving sub-200ms response time.
- Automated SOS alerts via Firebase Cloud Messaging with push notifications delivered within 2 seconds of threat detection.

Multi-Agent Research Assistant May 2026 – Aug 2026

Agentic AI | LangChain | GPT-4o | RAG | FastAPI | Next.js | [GitHub](#)

- Architected a 6-agent AI pipeline using LangChain + GPT-4o that autonomously orchestrates web research, RAG-based summarization, fact-checking, and multilingual PDF report generation, delivering cited reports in <60 seconds.
- Built a semantic retrieval layer using Pinecone + OpenAI embeddings (1536-dimensional), indexing 50+ chunks per session for multi-source knowledge grounding and confidence scoring.
- Reduced pipeline latency 53% (72s → 34s) using `asyncio.gather()` for parallel execution.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript.

AI/ML: Machine Learning, HuggingFace Transformers, LangChain, OpenAI API, PyTorch, TensorFlow, Scikit-learn, OpenCV, Streamlit.

Key Concepts: LLM Fine-Tuning, RAG Pipelines, Agentic AI, Prompt Engineering, NLP, Computer Vision, Deep Learning, ML Model Evaluation.

Data: NumPy, Pandas, Matplotlib, Data Visualization.

Databases: Pinecone (vector embeddings), Supabase, Firebase.

Web Frameworks: FastAPI, Flask, Django, Next.js, MERN Stack.

Cloud & DevOps: GCP (Cloud Run, Vertex AI), Docker, CI/CD, GitHub Actions.

Tools: Git, Jupyter Notebook, VS Code, Postman.

EDUCATION

B.Tech. in Computer Science and Engineering *Bareilly, India*
SRMS College of Engineering, Technology & Research | CGPA: 7.2/10 | Expected May 2027 2023 – Present

ACHIEVEMENTS & CERTIFICATIONS

- Naukri Young Turks 2025** – Ranked **428 nationally** out of thousands in a national-level technical assessment.
- DSA** – Solved **700+ problems** on LeetCode, CodeChef, and HackerRank; consistently top percentile rankings.
- IBM SkillsBuild** – Certified in *Introduction to RAG & Ethical Considerations of Generative AI*.
- Tata Group (Forage)** – Completed simulations in Cybersecurity Analysis, GenAI Analytics, and Data Visualisation.